OOP Handout 1 : Factors of Software Quality

- **What is a good program? (Factors of Good Software)**
    - correct (correctness)
        - satisfy user requirement/specification without error
    - efficient (efficiency)
        - optimizes resource usage to deliver fast response times
        - space / time , HW utilization
    - easy to maintain (maintainability)
        - easy to read (readability)
        - easy to modify (modifiability)
        - easy to extend(extensibility) , scalability
            - scalability: handles growing volumes of users, data, or concurrent tasks smoothly by adding infrastructure resources.
        - easy to reuse (reusability)
        - easy to debug , easy to test (testability)
        - modular (modularity)
        - easy to port (portability) , platform-independent (HW, OS, …)
    - stable (stability), reliable (reliability) , robust (robustness)
    - easy to use (usability) , easy to run
    - secure (security) , integrity (accurate & unaltered data, protection)
        - Protects data against unauthorized access, corruption, or malicious exploits.
    - Interoperability
        - Easily exchanges data and integrates with other software systems via standardized protocols and interfaces.
    - concise (conciseness)
        - writing code, designing architectures, and crafting interfaces that achieve maximum result with the minimum necessary complexity or syntax without sacrificing readability or correctness. (ex) minimized redundancy. simplified branch decisions
    - complete (completeness)
        - implements all required functionality
    - consistent (consistency)
        - Uniformity of design, behavior, coding standards, and user interfaces across an entire system.
        - ex) uniform naming conventions (e.g., camelCase vs snake_case), formatting
    - well-documented (documentation)
    - well structured